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COURSE ASSESSMENT 2025 COURSE DETAILS

Course Code:	WAD621s	Semester:	2
Course Title:	Web Application Development		
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COURSEWORK DETAILS

Title of Assessment:	University Timetable Manager
Formats:	Documentation
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Due Date:	26 September 2025

FULL-TIME SOFTWARE DEVELOPMENT GROUP

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1. Introduction & Background

The modern university experience presents students with increasingly complex scheduling challenges. Developing an ideal class schedule has grown to be a major administrative load for students due to the variety of courses, elective possibilities, campus locations and conflicting time slots. In order to maintain their schedules, many students now use manual techniques like spreadsheets, paper planners or screenshot collections, which results in inefficiencies, scheduling conflicts and missed opportunities for improved time management.

By offering a consolidated, interactive web tool that streamlines the whole scheduling process, the University Timetable Manager meets this urgent requirement. In the context of our rapidly digitizing educational environment, having a dependable digital tool for managing schedules is now essential. This project acknowledges that academic performance and efficient time management directly correlates, and that students may concentrate more on their studies by lowering the administrative burden of scheduling.

This project is important in ways that goes beyond practicality. It signifies a move toward intelligent, data-driven academic planning that may assist students in making the most of their time at university, balancing their workload more efficiently and choosing courses based on a visual time representation.

2. Problem Statement / Objectives

2.1. Problem to be Solved

There are currently minimal integrated, user-friendly tools available to students for efficiently creating, visualizing and managing their university schedules. The current manual methods are prone to mistakes, difficult to modify and don't provide features for conflict detection or schedule optimization. In addition, students can't easily share schedules with peers for coordination or save multiple schedule variations for comparison and cooperation.

2.2. Objectives

To develop a comprehensive web-based timetable management system that uses an easy-to-use drag-and-drop interface with local and cloud storage capabilities to let college students effectively create, modify, save, and manage their academic timetables.

- Implement an interactive drag-and-drop interface that lets students assign classes to timeslots while receiving visible feedback in real time.
- Create an automated conflict detection system that notifies users when they try to plan classes that overlap.
- Build a robust data persistence system with PHP/MySQL for secure, cross-device synchronization using Local Storage for quick and easy access.
- Design a mobile-friendly, responsive UI that functions flawlessly on smartphones, tablets and PC's.
- Build user authentication and profile management features to allow for the storage and retrieval of personalized schedules.
- Include features that let students share their schedules with peers to allow for coordination.
- Allow students to create and compare multiple schedule versions and manage them, enabling students to create and compare different scheduling scenarios.
- Provide functionality to export schedules for offline usage in a variety of formats.

3. Proposed Solution

The University Timetable Manager will be a full-stack web application that transforms how students manage their academic timetables. The application will feature a visually appealing grid-based interface representing the weekly calendar, where each cell corresponds to a specific time slot and day combination.

Core Features and Functionality

3.1. Interactive Schedule Builder:

The main component of the application will be a drag-and-drop interface where students can browse available courses from a sidebar panel and drag them directly onto their timetable grid. Instant visual feedback will be provided by the system which will also automatically avoid conflicts and identify acceptable drop zones. When a course is dropped, it will snap into place and display essential information including course code, name, location and lecturer.

3.2. Intelligent Conflict Management:

The application will implement conflict detection that checks for time overlaps, venue conflicts and pre-requisites. When conflicts are detected, the system will provide clear visual indicators and suggest alternative time slots or sections where available.

3.3. Multilayered Data Storage:

The solution will employ dual storage strategy. Local Storage will provide immediate, offline-capable storage for quick access and temporary saves. For permanent storage and cross-device synchronization, the backend will utilize PHP with MySQL database, implementing secure API endpoints for CRUD operations on schedule data.

Technical Architecture:

Frontend Technologies:

- HTML5 for semantic structure with accessibility-compliant markup
- CSS3 with Flexbox/Grid for responsive layouts
- Vanilla JavaScript for core interactivity, drag-and-drop functionality and DOM manipulation
- AJAX/Fetch API for asynchronous server communication
- Web Storage API for LocalStorage implementation

Backend Technologies:

- PHP 7.4+ for server-side logic and API development
- MySQL for relational database management
- RESTful API design for clean separation of concerns
- JSON for data exchange between frontend and backend
- Session management for user authentication

Additional Tools:

- Bootstrap or custom CSS framework for consistent UI components
- Chart.js for potential schedule analytics visualization
- PDF generation library for export functionality

4. Scope & Limitations

4.1. Within Scope

- Complete timetable creation and editing functionality with drag-and-drop interface
- User registration, login and profile management system.
- Course database with search and filter capabilities
- Schedule validation and conflict detection
- Multiple schedule versions per user account
- LocalStorage for client-side persistence
- MySQL database for server-side storage
- Responsive design for mobile and desktop browsers
- Basic schedule sharing via unique URLs
- Export functionality for common formats

4.2. Outside Scope

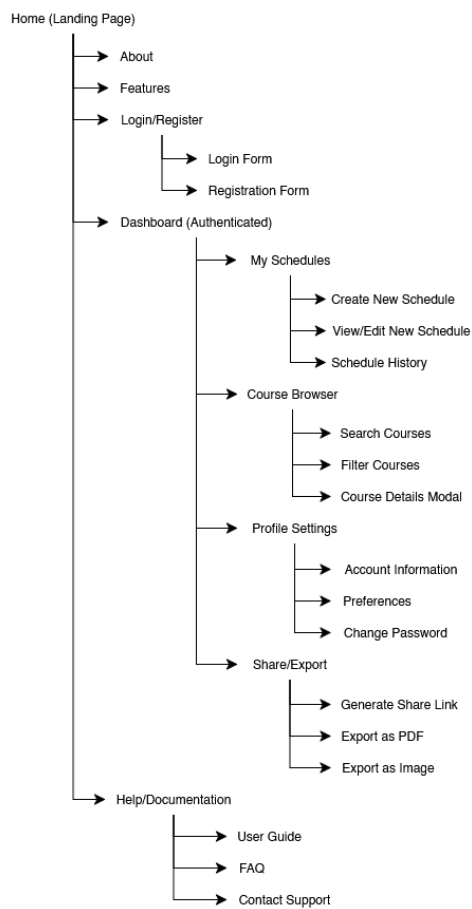
- Integration with actual NUST registration systems
- Automatic course enrolment or registration
- Real-Time collaboration features (multiple users editing simultaneously)
- AI-based schedule optimisation
- Native mobile applications
- Integration with third-party calendar applications beyond basic export
- Instructor/Admin interfaces
- Grade tracking/academic progress monitoring

4.3. Limitations and Constraints

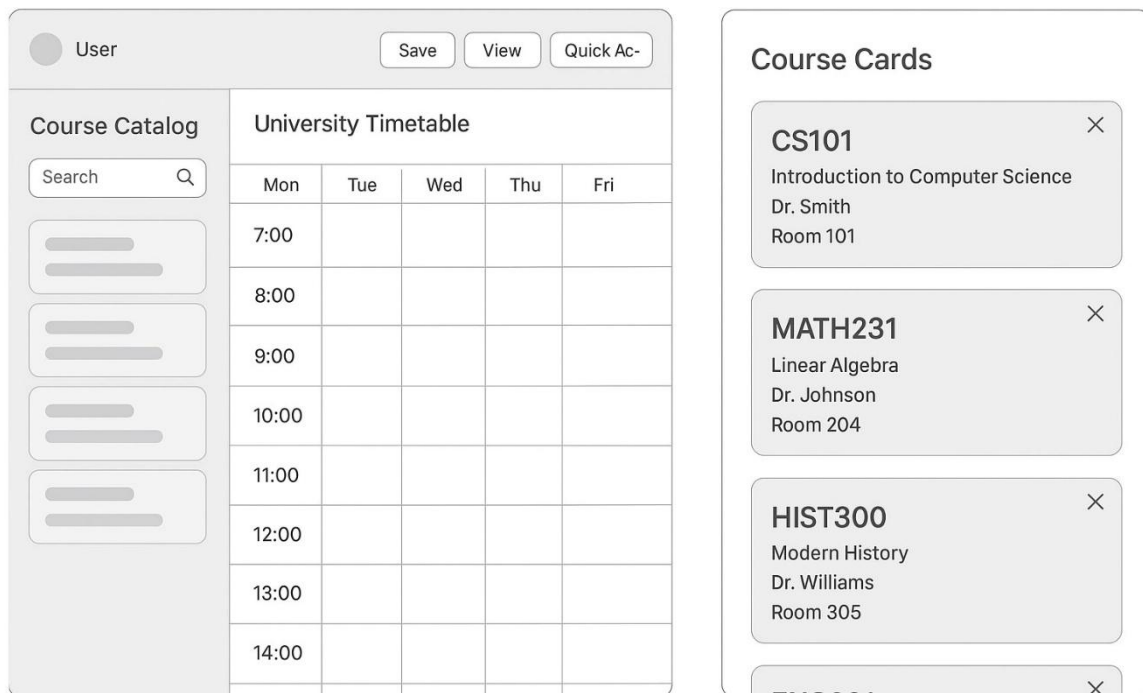
- Development timeframe limited to semester schedule
- Team size constraints may limit feature complexity
- No access to actual university APIs, requiring manual data entry or mock data
- Browser compatibility limited to modern browsers
- Performance optimization for schedule with more than 10 courses may be limited
- Security features will be basic, suitable for demonstration but not production deployment.

5. System Design

Site Map:



Wireframes and UI Concepts



Main Timetable Interface

The primary interface will feature a large grid of the screen with time slots on the Y0axis and the weekdays on the X-axis. A collapsible sidebar will contain the course catalogue with search functionality. The top navigation will include user menu, save button, view options and quick actions.

Course Cards Design:

Each course will be represented as a coloured card displaying course code prominently, with course name, instructor and room number in smaller text. Colours will be automatically assigned but customizable with a small "x" button for removal.

Navigation Flow

1. Entry Flow:

Landing page ⇒ Login/Register ⇒ Dashboard

2. Schedule Creation Flow:

Dashboard ⇒ Create New ⇒ Timetable Editor ⇒ Add Courses ⇒ Save ⇒ View/Share

3. Edit Flow:

My Schedules ⇒ Select Schedule ⇒ Edit Mode ⇒ Modify ⇒ Save Changes

4. Share Flow:

Open Schedule ⇒ Share Button ⇒ Generate Link ⇒ Copy/Send Link

6. Target Audience / Users

The main user group consists of undergraduate students who need to plan their semester schedules. These users are typically tech-savvy, familiar with web applications & value efficiency in their academic planning. They will benefit from reduced time spent on schedule planning, fewer scheduling conflicts & better visualization of their weekly commitments.

User Benefits

- Time saving schedule creation compared to manual schedule creation
- Elimination of scheduling conflicts through automatic detection
- Improved decision-making through visual schedule representation
- Enhanced collaboration with peers through schedule sharing
- Reduced stress during registration periods
- Better work-life balance through optimized time management
- Accessibility to schedules from any device with internet connection

7. Methodology / Development plan

The project will follow an Agile development methodology with iterative prototyping. This approach allows for continuous feedback integration and ensures that each feature is thoroughly tested before moving to the next. The development will be organized into sprints with specific deliverables for each sprint aligning with the course checkpoints.

- Daily Standups: Brief daily check-ins will ensure continuous communication about progress, blockers and needs for collaboration.
- Iterative Development: Features will be developed in iterations, starting with basic functionality and progressively adding complexity. Each iteration will include development, testing and refinement.
- Code Reviews: All code will undergo peer review before merging to ensure quality, consistency and knowledge sharing among team members.

Tools & Technologies

Development Environment:

- Visual Studio Code as the primary IDE with extensions for PHP, JavaScript & MySQL
- PHP for database management during development
- Git & Github for version control

Design & Prototyping:

- Draw.io for system architecture diagrams
- Figma/Adobe XD/Canva for creating mock-ups & prototypes

Testing & Quality Assurance:

- Manual testing with detailed test cases
- Multiple browser testing (Chrome, Firefox, Edge)

Documentation:

- Markdown for README and technical documentation
- Microsoft Word/Google Document to document technical documentation

8. Timeline

Day Range	Phase	Tasks	Deliverables	Checkpoints
Day 1-2	Project Initiation & Requirements	- Team formation - Proposal writing - Initial research & technology selection - Requirement gathering - User stories & database schema - API planning	Project Proposal & Requirements Spec	Checkpoint1: Proposal & Requirements Submission
Day 3-4	Design Phase	- Wireframes - HTML structure - Basic CSS & Navigation - Technical architecture (system, DB, backend setup) - API structure design	Static Prototype & Tech Design	
Day 5-6	Frontend Development	- Complete HTML/CSS - Responsive design - UI components & interactions - Cross-browser testing	Stylized Static Website	
Day 7-8	Backend Development	- PHP/API development - User authentication - Session management - Database connection	Basic Backend System	Checkpoint 2: Backend system Ready
Day 9-10	Integration	- Connect frontend & backend - AJAX & data sync - Advanced features implementation	Integrated System	
Day 11	Testing & Refinement	- Bug fixes- Performance optimization- Security review	Tested Application	Checkpoint 3: Tested System
Day 12	Polish & Accessibility	- UI improvements - Accessibility features - Final styling & browser checks	Polish Application	
Day 13	Documentation	- Technical docs - User documentation - Deployment preparation	Complete Project Package	
Day 14	Final Presentations	- Project demonstration - Q&A- Final submission	Sunmissions	Final Presentation & Submissions

9. Expected Outcomes

Functional Deliverables

1. Fully Functional Web Application
 - A complete, deployed University Timetable Manager with all core features operational, including drag-and-drop scheduling, conflict detection and data persistence.
2. Comprehensive Documentation
 - User manual, technical documentation
3. Database System
 - Fully normalized MySQL database with tables for users, courses, schedules and relationships, capable of handling multiple concurrent users.
4. Responsive User Interface
 - Cross-platform compatible interface that works seamlessly on devices from mobile screens to desktop displays

Technical Skills Gained

- Proficiency in full-stack web development using HTML, CSS3, JavaScript, PHP and MySQL
- Experience with modern development workflows, including version control, code review and agile methodologies
- Understanding of RESTful API design and implementation
- Knowledge of web security best practices including input validation
- Expertise in responsive web design and cross-browser compatibility
- Database design and optimization skills

Academic Learning Outcomes

- Application of theoretical web development concepts to practical , real-world problems.
- Project management experience including timeline management, resource allocation and risk mitigation
- Team collaboration and communication skills essential for professional development
- Problem-solving abilities through debugging and feature implementation
- Understanding of user experience principles and accessibility standards.
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Potential Long-term Impact

- The application could potentially be refined and adopted by the university for official use, benefiting thousands of students
- Portfolio enhancement by a substantial project demonstrating full-stack capabilities for future employment opportunities
- The project could be released as open source, allowing other institutions to adapt it for their needs.
- The codebase could serve as a starting point for more advanced features like AI-powered optimization or integration with university systems
- With further development, this could evolve into a commercial product for educational institutions.

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